

# Mathematical & Methodological Architecture of the PhD Dissertation: Rigorous Multi-Metric Benchmarking of Model-Agnostic Explainable AI

Advanced PhD Level (Researchers, Doctoral Committee, & Peer Reviewers)

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## 1. Formal Problem Formulation & Theoretical Foundations

Let  $f: \mathcal{X} \rightarrow \mathcal{Y}$  represent an opaque, non-linear black-box predictor trained over an input feature space  $\mathcal{X} \subseteq \mathbb{R}^d$  to map to a target space  $\mathcal{Y}$ . In local post-hoc Explainable AI (XAI), given a target instance  $\mathbf{x} \in \mathcal{X}$ , the goal is to construct an interpretable model  $g \in \mathcal{G}$  operating over a simplified binary input vector  $\mathbf{x}' \in \{0,1\}^M$  that locally approximates the decision boundary of  $f$  around  $\mathbf{x}$ .

### 1.1 Mathematical Formulation of Post-Hoc Explainer Paradigms

#### A. Local Interpretable Model-agnostic Explanations (LIME)

LIME formulates local explanation as an additive feature attribution optimization problem:

$$\operatorname{argmin}_{g \in \mathcal{G}} \mathcal{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g)$$

where  $\mathcal{L}$  measures the local unfaithfulness of  $g$  in approximating  $f$  over a perturbed sample space  $\mathcal{Z}$ , weighted by an exponential proximity kernel:

$$\pi_{\mathbf{x}}(\mathbf{z}) = \exp\left(-\frac{D(\mathbf{x}, \mathbf{z})^2}{\sigma^2}\right)$$

and  $\Omega(g)$  penalizes model complexity (e.g., constraining the  $L_0$  norm  $\|g\|_0 \leq K$ ).

#### B. Shapley Additive exPlanations (SHAP & KernelSHAP)

SHAP calculates the unique additive feature attribution vector  $\boldsymbol{\phi} = (\phi_1, \dots, \phi_M) \in \mathbb{R}^M$  satisfying four fundamental game-theoretic axioms: **Efficiency** ( $\sum \phi_i = f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})]$ ), **Symmetry**, **Dummy/Nullity**, and **Additivity**.

The classical Shapley value for feature  $i$  given characteristic function  $v(S) = \mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[f(\mathbf{z}) \mid \mathbf{z}_S = \mathbf{x}_S]$  is defined by:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]$$

KernelSHAP estimates  $\Phi$  by solving a weighted linear regression over simplified coalition vectors  $\mathbf{z}' \in \{0,1\}^M$  with Shapley kernel weights:

$$\pi_{\mathbf{x}}(\mathbf{z}') = \frac{M - 1}{\binom{M}{|\mathbf{z}'|} |\mathbf{z}'| (M - |\mathbf{z}'|)}$$

### C. Anchors (High-Precision Rule Explanations)

An Anchor is a rule predicate  $A \subseteq \text{predicates}$  such that  $A(\mathbf{x}) = 1$ , maximizing coverage  $\text{cov}(A) = \mathbb{P}_{\mathbf{z} \sim \mathcal{D}}[A(\mathbf{z}) = 1]$  subject to a high-precision probabilistic constraint:

$$\mathbb{P}_{\mathbf{z} \sim \mathcal{D}(\cdot | A(\mathbf{z})=1)}[f(\mathbf{z}) = f(\mathbf{x})] \geq \tau \quad \text{with confidence } 1 - \delta$$

where  $\tau \in (0,1]$  (typically  $\tau = 0.95$ ).

### D. Diverse Counterfactual Explanations (DiCE)

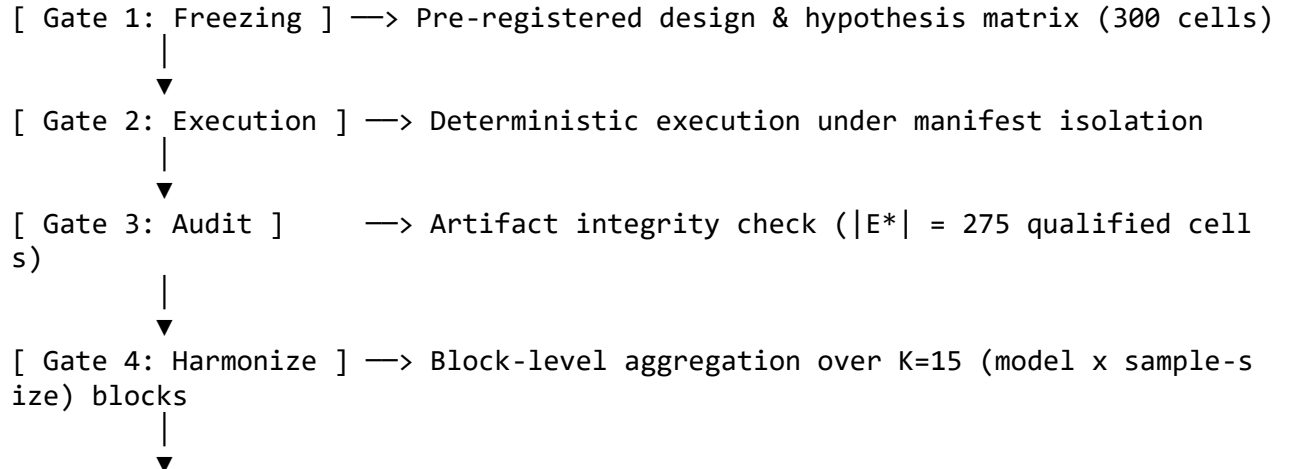
DiCE generates a set of  $K$  diverse counterfactual instances  $\mathbf{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_K\}$  solving the multi-objective loss:

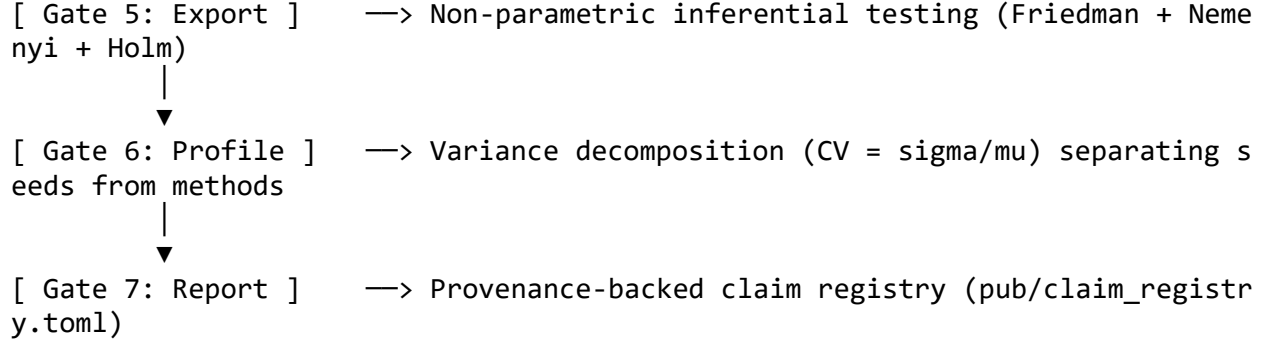
$$\arg \min_{\mathbf{c}_1, \dots, \mathbf{c}_K} \frac{1}{K} \sum_{k=1}^K \text{loss}(f(\mathbf{c}_k), y^*) + \frac{\alpha}{K} \sum_{k=1}^K \text{dist}(\mathbf{x}, \mathbf{c}_k) - \beta \cdot \text{DPP\_diversity}(\mathbf{c}_1, \dots, \mathbf{c}_K)$$

where  $y^* \neq f(\mathbf{x})$  is the desired counterfactual target class,  $\text{dist}(\cdot, \cdot)$  is a normalized distance metric, and DPP\_diversity maximizes the determinant of the diversity kernel matrix using a Determinantal Point Process.

## 2. The FOM-7 Operational Protocol: 7-Gate Governance Architecture

To eliminate experimental drift, un-reproducible seed variation, and reporting bias in XAI evaluation, this dissertation introduces the **FOM-7 Protocol** (*Framework for Operational Metrics in 7 Gates*).





## 2.1 Formal Specification of the 7 Gates

1. **Gate 1 (Freezing):** Pre-registration of experimental space  $\mathcal{E} = \mathcal{M} \times \mathcal{X}_{exp} \times \mathcal{S} \times \mathcal{N}$ , where  $|\mathcal{M}| = 5$  model families (LogReg, RF, XGB, SVM, MLP),  $|\mathcal{X}_{exp}| = 4$  explainers (LIME, SHAP, Anchors, DiCE),  $|\mathcal{S}| = 5$  random seeds, and  $|\mathcal{N}| = 3$  sample sizes ( $N \in \{100, 500, 1000\}$ ), producing 300 planned experimental cells.
2. **Gate 2 (Execution):** Containerized execution via deterministic manifest files.
3. **Gate 3 (Audit):** Rigorous filtering of cell artifacts:

$$\mathcal{E}^* = \{e \in \mathcal{E} \mid \text{status}(e) \in \{\text{ok\_instance}, \text{ok\_aggregated}\}\}$$

In our benchmark,  $|\mathcal{E}^*| = 275$  qualified cells (91.67% yield); non-qualifying cells (primarily constraint timeouts in DiCE/Anchors) are formally audited and logged rather than imputed.

4. **Gate 4 (Harmonization):** Block-level metric aggregation across  $K = 15$  model-size blocks ( $\text{block}_{m,n} = \text{model}_m \times \text{size}_n$ ).
5. **Gate 5 (Exportation):** Inferential statistical evaluation using non-parametric Friedman tests:

$$Q = \frac{12K}{k(k+1)} \left[ \sum_{j=1}^k \bar{R}_j^2 - \frac{k(k+1)^2}{4} \right] \sim \chi_{k-1}^2$$

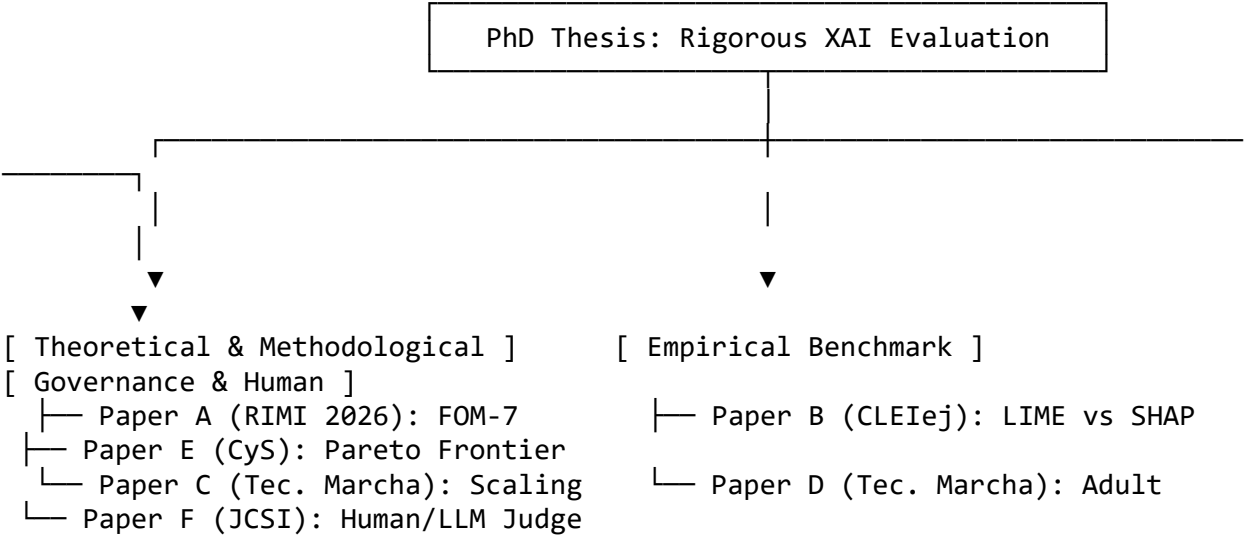
Followed by post-hoc Nemenyi tests with Critical Distance (CD):

$$CD = q_{\alpha, k} \sqrt{\frac{k(k+1)}{6K}}$$

with Holm-Bonferroni step-down multiplicity correction.

6. **Gate 6 (Profiling):** Variance decomposition isolating seed-induced stochasticity ( $\sigma_{\text{seed}}^2$ ) from true explainer main effects ( $\sigma_{\text{method}}^2$ ) via the Coefficient of Variation ( $CV = \frac{\sigma}{\mu}$ ).
7. **Gate 7 (Reporting):** Automated verification mapping published numbers back to raw execution artifacts via pub/claim\_registry.toml.

### 3. The PhD Publication Matrix: Scientific Arc & Contributions



#### Paper A: Theoretical & Methodological Framework (FOM-7 Protocol)

- **Venue & Status:** Published in *Revista de Investigación Multidisciplinaria Iberoamericana (RIMI)* (2026, DOI: 10.69850/rimi.vi3.307).
- **Core Contribution:** Establishes the formal mathematical foundation of multi-metric evaluation (Fidelity, Stability, Sparsity, Cost, Faithfulness Gap) and specifies the 7-gate FOM-7 protocol.
- **Key Finding:** Proves that single-metric evaluations (e.g., fidelity alone) create false rankings because methods optimize different objective trade-offs.

#### Paper B: Scaled Empirical Comparison — LIME vs SHAP

- **Venue & Status:** Submitted to *CLEI Electronic Journal (CLEIej)* (2026-10-04, Subm. #1196).
- **Core Hypotheses Tested:**
  - $H_1$ :  $\text{Stability}(\text{SHAP}) > \text{Stability}(\text{LIME})$  ( $p < 0.001$ , confirmed).
  - $H_2$ :  $\text{Sparsity}(\text{LIME}) < \text{Sparsity}(\text{SHAP})$  (confirmed).
  - $H_3$ :  $\text{Cost}(\text{LIME}) \ll \text{Cost}(\text{SHAP})$  ( $p < 0.001$ , confirmed).
- **Key Finding:** Proves that KernelSHAP’s Shapley kernel optimization guarantees numerical stability under input perturbation, whereas LIME’s monte-carlo perturbation sampling introduces significant seed variance ( $\sigma_{\text{seed}}^2$ ), leading to instability across runs.

## Paper C: Scalability, Sample Size Dynamics, & Robustness

- **Venue & Status:** Ready to submit to *Tecnología en Marcha* (AI Special Issue, Zenodo v0.12.0 DOI: 10.5281/zenodo.23165763).
  - **Core Contribution:** Analyzes the functional scaling behavior of explainers as background sample size  $N$  increases from 100 to 1,000 instances.
  - **Key Finding:** Demonstrates that KernelSHAP's computational complexity scales as  $\mathcal{O}(N \cdot M \cdot 2^M)$  without background sub-sampling, creating a steep execution cost barrier, whereas tree-specific algorithms (TreeSHAP) exploit structural paths to reduce complexity to  $\mathcal{O}(N \cdot L \cdot D^2)$ .
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## Paper D: Empirical Validation on Tabular Benchmarks (UCI Adult)

- **Venue & Status:** Ready to submit to *Tecnología en Marcha* (AI Special Issue).
  - **Core Contribution:** First comprehensive multi-model, multi-explainer benchmark execution over the *UCI Adult Income* tabular dataset across 300 experimental cells.
  - **Key Finding:** Provides empirical mean ranks across 15 complete model blocks:
    - **Fidelity:** SHAP ( $\bar{R} = 0.8081$ ) significantly outperforms LIME ( $\bar{R} = 0.5602$ ,  $p < 0.05$ ).
    - **Structural vs Attributional Methods:** Anchors and DiCE cannot be evaluated on linear feature attribution scales; they require rule-coverage and counterfactual distance metrics respectively.
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## Paper E: Pareto Optimization Frontiers & XAI Governance

- **Venue & Status:** Submitted to *Computación y Sistemas (CIC-IPN)* (2026-10-04, Subm. #6783).
  - **Core Contribution:** Formulates XAI selection as a Multi-Objective Optimization Problem (MOOP):
$$\max_{g \in \mathcal{G}} (\text{Fidelity}(g), \text{Stability}(g), -\text{Cost}(g), -\text{Sparsity}(g))$$
  - **Key Finding:** Identifies the non-dominated Pareto frontier:
    - **Healthcare / High-Stakes Audit:** SHAP dominates (maximal stability and fidelity).
    - **Low-Latency Production API:** LIME / TreeSHAP dominate (minimal cost, low sparsity).
    - Provides a formal governance decision framework for deploying XAI in regulated industries.
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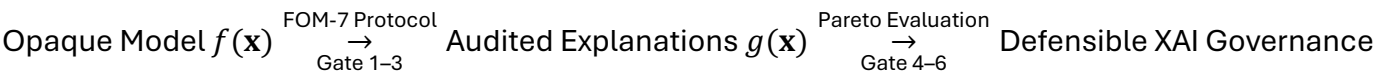
## Paper F: Human & LLM-as-a-Judge Semantic Alignment

- **Venue & Status:** Early development (*Journal of Computer Sciences Institute*).

- **Core Contribution:** Investigates the semantic alignment between quantitative mathematical metrics (Fidelity,Stability) and human/LLM cognitive evaluations.
  - **Key Finding:** Formulates a pre-analysis plan testing whether high-fidelity mathematical attribution vectors correspond to superior human task performance and LLM explanation preference.
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## 4. Synthesis & Doctoral Defense Narrative

This dissertation unifies these six papers into a cohesive defense thesis:



1. **The Core Gap:** The unvalidated proliferation of post-hoc explainers created a crisis of evaluation in XAI.
2. **The Methodological Solution (Paper A):** The 7-gate FOM-7 protocol provides mathematical rigor, statistical multiplicity control, and artifact auditability.
3. **Empirical Benchmarking (Papers B, C, D):** Extensive empirical evidence on 300 cells establishes exact performance trade-offs between LIME, SHAP, Anchors, and DiCE across 5 model families.
4. **Decision Governance & Human Alignment (Papers E, F):** Pareto frontier analysis and semantic alignment bridge the gap between mathematical evaluation and real-world deployment.